



RAMCO INSTITUTE OF TECHNOLOGY

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Department of Computer Science and Engineering

Academic Year 2023 – 2024 (Odd Semester)

Degree, Semester & Branch: III Semester B.E. CSE 'A'

Course Code & Title: CS3391 Object Oriented Programming

Name of the Faculty member (s): Mrs.B.Vijayalakshmi, AP (SG)/CSE

Innovative Practice Description

- Unit / Topic: Unit V / JavaFX Controls

- Course Outcome: CO5

- Topic Learning Outcome: TLO11

- Activity Chosen: Design Activity

- Justification:

The JavaFX UI controls is used in designing GUI for an application. The understanding of the design part is essential for developing an application front end in a very good manner. Hence to improve the knowledge on various UI controls the design activity is chosen which also helps to assess the understanding level of the students.

- Time Allotted for the Activity: 15 Minutes

- Details of the Implementation:

- The user interface design is an interesting activity in which the students are given a simple application to design.
- The various UI controls in JavaFX was taught to the students.
- At the end of the session 15 minutes were given to them to complete the given design.
- The design consists of a set of various UI controls in which the corresponding code has to be completed by the students.
- The design code was evaluated by the instructor and identified the places where the students get need more clarity.
- The sample responses of the code designed by the students are shown in Fig.1.

- CO – PO / PSO mapping:

CO	PO1	PO2	PO3	PO5	PO8	PO12	PSO1	PSO2	PSO3
CO5	3	3	2	1	1	1	3	1	1

(1 – Low 2 – Moderate 3 – High)

- PO / PSO mapped:

Innovative practice	PO1	PO2	PO3	PO5	PO9	PSO1	PSO2	PSO3
	3	2		1	1	3	1	1

Justification for correlation	The students apply the knowledge of JavaFX UI	The students analyze and write design code for the design.		The students design the given UI using JavaFX controls.	The students individually write their design code.	The JavaFX is used in developing the GUI.	The concept is used in providing Cloud Computing based solutions.	The concept is used in developing interactive Java applications to providing AI based solutions
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• **Images / Screenshot of the practice:**

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Count Increment

```

class CountApp extends Application {
    public void start (Stage stage)
    {
        Label l1 = new Label ("Counter:");
        TextField t1 = new TextField ();
        Button b1 = new Button ("Count");

        HBox hb = new HBox ();
        hb.getChildren ().addAll (l1, t1, b1);

        GridPane grid = new GridPane ();
        grid.setAlignment (CENTER);
        grid.add (hb);

        b1.setOnAction (event -> countNumber ());

        Scene s1 = new Scene (grid);
        Stage -> setScene (s1);
        Stage.show ();
    }
  
```

```

public class SampleTest extends Application
{
    public void start (Stage stage)
    {
        Label l1 = new Label ("Counter");
        TextBox tb1 = new TextBox ();
        tb1.setValue ("0");
        Button b1 = new Button ("Count");
        FlowPane fp = new FlowPane ();
        fp.getChildren().addAll (l1, tb1, b1);
        fp.setHGap (20);

        Stage
        Scene s1 = new Scene (fp, 500, 480);

        Stage.setTitle ("JavaFX Counter");
        Stage.setScene (s1);
        Stage.show ();
    }
}

```

Fig. 1: Sample Design code for a counter application

- **Reflective Critique:**

- ❖ **Feedback of practice from students and other stakeholders:**

- Most of the students are interested in writing the code for the given design. They make use of the learnt UI controls in the page design

- ❖ **Benefit of the practice:** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

- This activity provides a way to exhibit their creativity while designing.
 - The students' understanding level can be assessed with their design code. Those who need more clarification are identified and given more clarity in the next session.

- ❖ **Challenges faced in implementation:**

- Some of the students struggled to design the page and it takes more time to evaluate design of all the students.

References:

- <https://www.monash.edu/learning-teaching/TeachHQ/Teaching-practices/Active-learning/active-learning/design-active-learning>